

Mapping "Cult," Second Pass: The v2 Geometric Analysis

Working report – draft base material for the Results chapter,
with a first attempt at interpretation

Prepared for Célestin Meunier's thesis project

Covers run 20260909-224215, generated 2026-09-10

How to read this document. This is still not a finished Results chapter, and the descriptive sections below (Sections 2–7) report numbers, not arguments, exactly as the predecessor v1 document did. What is new here, at the researcher's request, is Section 8: a first, explicitly provisional pass at what these patterns *might* mean. That section is kept structurally separate and visually marked (a tan box, like this one is blue) throughout, precisely so a reader can tell at a glance which sentences are geometric fact and which are a reading of that fact that the researcher is free to revise, argue with, or reject outright before anything becomes actual thesis prose. Nowhere in this document is geometric proximity treated as evidence that any group, practice, or person *is* or *is not* "a cult" – that distinction is kept explicit everywhere it matters.

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1 In short: what did this run find?

For a reader who wants the gist before the detail. Every claim below is unpacked, with numbers and caveats, later in this document.

- **The corpus got much smaller, on purpose.** A manual fidelity check of the first extraction pipeline (v1) found it was pulling out paraphrases, mistranslations, and non-claims (headings, citations) as if they were genuine statements about cults. The pipeline was rebuilt around a strict "must be an exact quote" rule plus a second model that independently checks every candidate. The result: far fewer expressions survive (literature alone dropped from 39,236 to 5,741 – an 85% cut), but the ones that remain are verbatim and checked. This document is about that smaller, stricter dataset.
- **The three sources still barely form separate clusters** – same conclusion as v1's run, at a similarly-near-zero silhouette score. Literature, the French government's MIVILUDES reports, and interview responses overlap substantially rather than occupying clearly separate regions of the space.
- **But something changed in how the three sources relate to each other.** In v1, the three corpora's average positions pointed in almost unrelated directions (cosine similarities near zero). In v2, literature, MIVILUDES, and interviews all point in a *noticeably similar* direction (cosine similarities of 0.52–0.64). One plausible reading, spelled out with its own caveats in Section 8, is that stripping out noise (paraphrase drift, boilerplate, mislabelled non-claims) left a cleaner signal of what these three sources have in common when they talk about cults – but a much smaller sample size can itself produce a tidier-looking average almost by chance, so this is not read as settled here.

- **Named things (people, specific groups) tend to sit at the edge of their own corpus's "typical" region; generic statements about cults/sects sit near the centre.** A new module run for the first time this session (Section 6.7) found this pattern in all three corpora.
- **One genuine, small data-quality finding:** a 97-year-old interviewee's transcribed answer, "Sept?", is almost certainly a mishearing of the French word *secte* (cult) as *sept* (seven) – not a substantive answer – yet it passed every automated quality check. It is one point out of 10,646 in the whole space; not large enough to change any result, but worth being honest about (Section 5).
- **The interviews corpus is now so small (64 expressions) that no single named entity is mentioned by all three corpora at once** – a real structural fact about this run's data, not a bug (Section 6.6).

2 Where the project stands

2.1 Why a second pass at all

A 100-item manual fidelity review of the first extraction pipeline (2026-09-08) found it systematically over-inclusive: expressions that paraphrased, mistranslated, or truncated their source; document headings and bibliography citations pulled out as if they were claims; occasional interviewer questions mislabelled as participant statements. The pipeline (**v2**) was rebuilt around a strict verbatim-span rule – the extraction model may only return an exact substring of the source text, checked by a deterministic code layer, not another model call – plus a Stage-1 text-integrity audit for broken PDF character-encoding, deterministic (not model-guessed) interview speaker-turn attribution, and a second, independently-prompted local model that judges every retained candidate against the same criteria a human reviewer would use. This "judge" model is described everywhere in this document as *model-judged*, never as a substitute for human review. Full design rationale is in `thesis/03_Content/3_Methods.tex`, section "Revising the Extraction Pipeline"; v1's own archives, shared space, and every retained run are left untouched on disk as a frozen historical comparison point, not deleted or overwritten.

2.2 Stage 1 – The same three corpora, re-extracted

The underlying source documents are the same three corpora as v1 (57 scholarly works, 2 MIVILUDES documents, 26 interview transcripts) – nothing was added or removed at the document level. What changed is how expressions were pulled out of them (previous subsection).

2.3 Stage 2 – A much smaller, stricter set of expressions

Running the v2 pipeline end-to-end on all three corpora produced:

Corpus	v1 expressions	v2 expressions	Change
literature	39,236	5,741	–85.4%
MIVILUDES	914	108	–88.2%
interviews	230	64	–72.2%

Table 1: Expression counts, v1's over-inclusive pipeline versus v2's strict verbatim pipeline, same underlying documents.

This is not a case of "different data" – it is the same documents, screened far more strictly. Section 5 reports this session's own quality check on a random sample of the v2 output.

2.4 Stage 3 – The same three reference vocabularies

The three background word-lists (concept backbone, structural concepts, ConceptNet-generalised concepts) are unchanged from v1 – they were built independently of either extraction pipeline and are reused as-is (see the v1 document, Section 2, for how each was built).

2.5 Stage 4 – Far fewer emergent entities

With a much smaller corpus, far fewer named things are mentioned often enough to count as an **emergent entity**. v1 required at least 3 mentions and still found 3,251 entities; applying the same threshold to v2 would have left only 1 entity, so the threshold was lowered to **1 mention** for this run – an explicit, logged design decision, not a silent change. This produced **21** emergent entities (see Section 6.6).

2.6 Stage 5 – One shared geometric space, rebuilt

The same pooling procedure as v1 (embed everything with the same model, standardize, then PCA down to the smallest dimensionality that keeps 95% of the variance) was re-run on the v2 expressions plus the 17 MIVILUDES criteria and the three (unchanged) reference vocabularies. This left **10,646** points, compressed to **389** dimensions (95.0% variance kept) – a much smaller space than v1's 44,520 points in 394 dimensions, but built the same way.

2.7 Stage 6 – The interview "first impression" layer: ported, not redone

25 of the 26 interviews' manually-verified opening-prompt exemplars no longer appear verbatim in v2's much stricter extraction (checked directly). Rather than lowering standards to force them back in, the existing, already-human-verified selections and their embeddings were kept exactly as they were and simply projected through the new v2 coordinate system instead of v1's. This is the same "port, don't redo" principle used elsewhere in this project: the embedding of a quote is a property of its own words, not of which pipeline version happened to be running when someone selected it.

2.8 Stage 7 – Rechecking whether the equal- n correction still makes sense

Because v2's corpus is so much smaller, it was worth explicitly rechecking whether the same correction used throughout v1 for literature's dominance still applies. It does: literature is **97.09%** of v2's expression points (versus 97.44% in v1 – essentially unchanged), so the same equal-size/equal- n bootstrap correction used throughout is still necessary and is used unchanged. What does change is the absolute size of that equal- n draw: v1 could draw 204 points per source (matching the smallest, interviews); v2 can only draw **64** (its own new smallest source). The correction logic is identical; the numbers it produces this run rest on a smaller, noisier sample, and confidence intervals reported below should be read with that in mind.

2.9 Stage 8 – Running the toolkit against v2, and a spot-check

The same reusable analysis toolkit used for v1 – now parameterized to point at either shared space via a single `-shared-space-dir` flag rather than hardcoded v1 paths – was rerun end-to-end against the v2 space (run 20260909-224215, covered by this document), plus a tenth module, `analyze_typicality`, run for the first time this session (Section 6.7). One module,

`analyze_emergent_entities`, encountered a genuinely empty result for its "entities mentioned by all three corpora" deliverable (Section 6.6) – fixed to log and skip that one deliverable explicitly rather than aborting the whole module, since an empty result here is a real fact about this run's small corpus, not an error to paper over. Separately, a fresh manual/automated spot-check was run on the retained v2 output itself (Section 5).

3 Key ideas, explained in plain language

Unchanged from the v1 document – included here again so this document is self-contained. Skip if already familiar with embeddings and dimensionality reduction.

3.1 Embeddings and vector spaces

An **embedding** is what you get when a piece of text is fed through a trained language model and converted into a long list of numbers – a **vector** (1,024 numbers, here). Texts with similar meaning end up as nearby lists of numbers: "a group led by a controlling figure" and "an authoritarian leader dominates the members" would land close together; the same phrase and "the recipe calls for two eggs" would land far apart. A collection of such vectors is a **vector space** or **embedding space**; because "nearby in the numbers" tracks "similar in meaning," geometric ideas – distance, direction, clustering – become tools for studying meaning.

3.2 Distance and similarity

- **Euclidean distance**: ordinary straight-line distance, just in many more than two dimensions. Smaller = more similar.
- **Cosine similarity**: whether two points point in the *same direction* from the origin, ignoring vector length. Ranges -1 (opposite) to 1 (identical direction), 0 meaning unrelated.

Both are always reported side by side, never collapsed into one number.

3.3 Centroid

The **centroid** of a group of points is their average position – a stand-in for "the centre of gravity" of, say, everything the literature corpus says, so whole-source comparisons reduce to one number (the distance between two centroids) instead of comparing every point pair.

3.4 Standardization and PCA (dimensionality reduction)

Every dimension of every vector was **standardized** before pooling, so no single source dominates purely because its raw numbers happen to be larger. **PCA** then compresses the 1,024-number description of each point down to the smallest number of dimensions that still keeps 95% of the original variation – 389 dimensions this run. Every distance and similarity number below is computed in this 389-dimensional space, not the original 1,024.

3.5 UMAP and 2-D pictures

389 dimensions cannot be drawn on a page. **UMAP** produces a readable (if approximate) 2-D layout purely for visualisation, trying to preserve which points were close/far in the real space. A plain 2-D **PCA slice** (the first two principal components) is shown alongside for the same reason two projections of a 3-D object each show something the other misses. Neither picture is itself evidence – the real numbers come from the full 389-dimensional space.

3.6 k-nearest-neighbour composition

A point's **k nearest neighbours** are the k closest other points ($k = 10, 20$ here). Averaging their source-composition over many points from a given source asks "how much do these sources actually overlap, versus sitting in their own separate regions?"

3.7 Silhouette score

A number, roughly -1 to 1 , scoring how well a proposed grouping (the three expression sources) matches the data's actual geometric clustering. Near 0 : the groups do not correspond to distinct clusters. Near 1 : they line up almost perfectly with the data's real shape.

3.8 Bootstrap resampling and confidence intervals

Figures reported as "mean, 95% CI [low, high]" come from **bootstrap resampling**: repeating a measurement many times (100 here) on randomly resampled subsets, then looking at how much the answer varies. A narrow interval means the measurement is stable; a wide one means it is noisier than it looks.

3.9 "Point role" versus "source dataset"

source_dataset: which of the eight underlying datasets a point came from. **point_role**: a coarser grouping into **expression** (an actual claim), **reference** (background vocabulary, not a claim), or **emergent** (a named thing the corpora mention). Figures are generally shown coloured both ways.

3.10 "Equal-size-controlled" and "equal- n " comparisons

The three sources are wildly different in size (5,741 / 108 / 64 this run), and the three reference vocabularies are too (3,000 / 1,500 / 195). A raw "which is nearest, on average" comparison is biased simply by pool size. An **equal-size-controlled** comparison down-samples every set to the smallest before comparing. This controls for *one* bias (size) only – not evidence the sets are otherwise interchangeable.

3.11 Typicality and "borderline" points (new this run)

A point's **typicality**, in this document, is nothing more than *how close it sits to its own group's centroid*. Close = "typical" of that group's average position in the space; far = "atypical." A **borderline** point, between two groups A and B , is one whose distance to A 's centroid and to B 's centroid are nearly equal – geometrically in-between, not evidence that the underlying claim is genuinely ambiguous or disputed in meaning. Both notions describe *this embedding space, under this model, this run* – not cognitive prototypicality or real-world category membership.

4 The datasets pooled into the shared space

A ninth, separate layer – the 25 ported interview prototypes (**point_role**="prototype") – sits in the same coordinate system but is not part of the 10,646-point count or the PCA fit itself (Stage 6 above).

Source	Points	Role	What it is
literature	5,741	expression	Claims/definitions, strict-verbatim extraction, from 57 scholarly works
miviludes	108	expression	Claims, strict-verbatim extraction, from 2 MIVILUDES documents
interviews	64	expression	Claims, strict-verbatim extraction, from 26 interview transcripts
miviludes_criteria	17	expression	MIVILUDES's own official 17 sectarian-drift criteria (unchanged from v1)
concept_backbone	3,000	reference	Topic-neutral general English concepts (unchanged from v1)
structural_concepts	1,500	reference	Generic structural/relational vocabulary (unchanged from v1)
conceptnet_concepts	195	reference	Hand-reviewed ConceptNet generalisation (unchanged from v1)
emergent_entities	21	emergent	Named things mentioned ≥ 1 time (threshold lowered from v1's 3; see Stage 4)

Table 2: The eight sources pooled into the 10,646-point, 389-dimensional v2 shared space.

5 Spot-check: is this run's data trustworthy?

Before writing up any result from this run, a fresh random sample was drawn from each corpus's final, judge-accepted output (100 literature, 40 MIVILUDES, 30 interview expressions; seed 42; full methodology and config in `processed/audits/v2_spotcheck_20260909.md` and the sample files themselves under each corpus's own `review/` folder). Two things were checked: an automated pass (does every sampled expression's text actually occur verbatim in its own recorded context; does the judge model's own faithfulness/extraction-issue flags line up with acceptance), and an independent read of a subset of rows by the researcher's assistant.

Automated pass: 0 of 170 sampled rows failed the verbatim check; 0 had a judge-flagged faithfulness or extraction-issue problem. This mostly confirms the judge model's own labels are internally consistent – expected, since the sampled file only contains judge-accepted rows – rather than an independent signal on its own.

Independent read: literature and MIVILUDES rows read as expected throughout – correctly attributed claims/definitions/named groups, sensible claim-mode and epistemic-status tags. One literature row's expression looked, at first glance, unrelated to the printed context; the full context window (an encyclopedia source, one entry ending and the next beginning mid-chunk) shows the expression is genuinely present verbatim – a reading artefact from truncating a long context window for a quick look, not an extraction error.

One genuine finding: "Sept?". v2's shared-space build deliberately does not filter by a minimum word count (`-min-expression-words 0`), because a plain word count cannot distinguish a meaningful short item ("Tomato cult!", "Illuminati", "Gourou") from a meaningless fragment. Checking every ≤ 2 -word expression across all three corpora (1,132 literature, 7 MIVILUDES, 17 interviews) confirms this worked as intended in all but one case: 16 of 17 short interview items are genuine named entities or associations.

The exception, interview `b3-aug23-1213` (participant, age 97): asked in French "*Quand je te dis le mot secte à quoi tu penses?*" ("When I say the word 'secte' [cult], what do you think of?"), the participant answered "**Sept?**" ("seven?") – almost certainly a mishearing of *secte* as *sept*, not a substantive response. The judge model scored it faithful, self-contained, cult-relevant, and atomic, correctly noting

it *is* a direct quote from the transcript – true, but the string itself carries no propositional content about cults, so it embeds wherever "Sept?" alone happens to land, not wherever a genuine response would. Scale: 1 of 10,646 points (0.01% of the whole space; 1.6% of the interviews corpus). Left in place, unmodified, as an explicit, logged limitation rather than a silent unilateral exclusion – consistent with how the `-min-expression-words/-entity-anchor-min-mentions` threshold choices elsewhere in this project were surfaced rather than decided silently.

6 What the analysis toolkit found

Ten modules were run against the v2 space in run 20260909-224215 (nine carried over from v1, plus `analyze_typicality`, new this session). Each subsection covers one module.

6.1 Sanity check: can interview extraction order stand in for what people think of first?

Module: `audit_free_listing_rank`

Same check as v1, same verdict: **not usable**. The narrowest reading – "rank 1 lands on the participant's own first claim, not the interviewer's question" – holds for 18 of 25 documents (72.0%), up from v1's 42.3% (11/26) but still far from a guarantee, and the underlying structural problem is unchanged: participants were asked for one example and then questioned about that same example, not asked to list several items in ranked order. `response_rank` plays no role anywhere else in this report; interview-side "first impression" evidence comes exclusively from the manually-ported initial-exemplar layer (Section 6.5).

6.2 Global structure: how far apart are the sources, overall?

Module: `analyze_global_structure`

Source A	Source B	v1		v2	
		Euclidean	Cosine	Euclidean	Cosine
literature	miviludes	7.18	−0.04	8.07	0.52
literature	interviews	9.97	0.04	8.19	0.64
miviludes	interviews	12.25	−0.06	12.07	0.26
literature	miviludes_criteria	13.94	−0.26	14.52	0.17
miviludes	miviludes_criteria	10.89	0.57	12.16	0.51
interviews	miviludes_criteria	17.64	−0.15	17.68	−0.02

Table 3: Selected pairwise centroid distances, v1 versus v2. Full 8×8 v2 matrix in `pairwise_source_centroid_matrix.csv`.

Euclidean distances barely moved. Cosine similarities moved a great deal, for every pair of the three expression corpora: from near-zero/slightly negative in v1 to solidly positive (0.26–0.64) in v2. In plain terms, the three corpora's *average positions* already sat about as far apart as before, but now point in noticeably more similar *directions* from the origin than they did under the old, noisier extraction. (What this might mean, and what it might not, is picked up in Section 8.)

Distance from each of the three reference vocabularies and the emergent-entities set to the combined-expression centroid ("full" variant; two other variants exist, never merged with

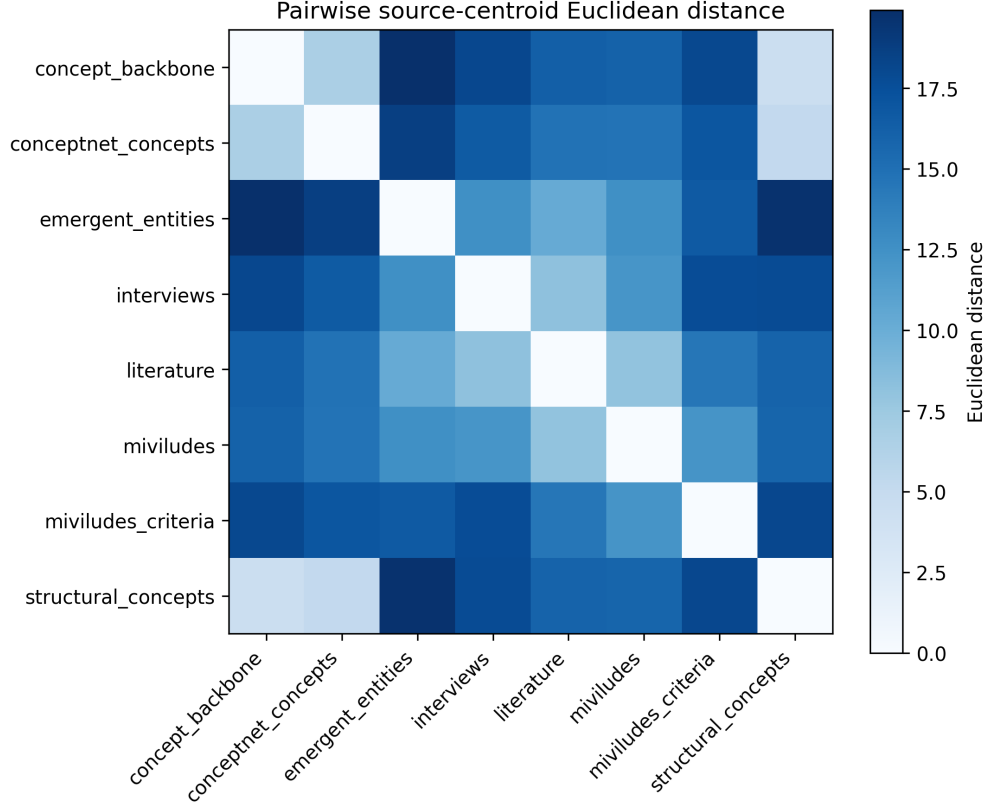


Figure 1: Heatmap of pairwise centroid distances across all eight v2 sources.

this one): concept_backbone 16.29 (v1: 16.70), structural_concepts 15.93 (v1: 16.18), conceptnet_concepts 14.80 (v1: 14.92), emergent_entities 10.22 (v1: 13.20). The ordering is unchanged from v1 (emergent entities closest, concept backbone furthest), but emergent_entities’ own distance shrank the most of the four – plausibly a direct consequence of the entity set itself shrinking from 3,251 to 21 (Section 6.6).

6.3 Cluster structure: do the three sources actually separate?

Module: analyze_cluster_structure

Same equal- n bootstrapped design as v1, now drawing 64 points per source (v1: 204) per Stage 7’s discussion above.

k	Query source	Neighbour source	v1 mean [95% CI]	v2 mean [95% CI]
10	literature	literature	0.316 [0.283, 0.350]	0.283 [0.225, 0.339]
10	literature	miviludes	0.307 [0.273, 0.338]	0.285 [0.230, 0.338]
10	literature	interviews	0.378 [0.347, 0.413]	0.432 [0.373, 0.495]
10	miviludes	literature	0.112 [0.089, 0.137]	0.136 [0.086, 0.173]
10	miviludes	miviludes	0.674 [0.641, 0.720]	0.613 [0.562, 0.670]
10	miviludes	interviews	0.213 [0.183, 0.241]	0.252 [0.210, 0.295]
10	interviews	literature	0.057 [0.043, 0.074]	0.156 [0.105, 0.213]
10	interviews	miviludes	0.088 [0.072, 0.104]	0.147 [0.113, 0.170]
10	interviews	interviews	0.855 [0.837, 0.875]	0.697 [0.644, 0.752]

Table 4: k-NN neighbour composition (equal- n , $k = 10$), v1 versus v2.

Every source is less self-clustering in v2 than in v1: interviews drop from 85.5% to 69.7% self-neighbours, MIVILUDES from 67.4% to 61.3%, literature from 31.6% to 28.3%. The $k = 20$ table (same CSV) shows the same pattern. Silhouette for the three-source grouping: **0.0130** (95% CI [0.0095, 0.0164]) – v1: 0.019 [0.017, 0.022]. Both are very close to zero; the conclusion "the three sources overlap substantially rather than forming three separate clusters" is unchanged from v1. Whether the *drop* in self-clustering reflects something real about the stricter extraction, or is simply noisier estimation from a much smaller equal- n draw (Stage 7), is taken up in Section 8 – it is not resolved by this table alone.

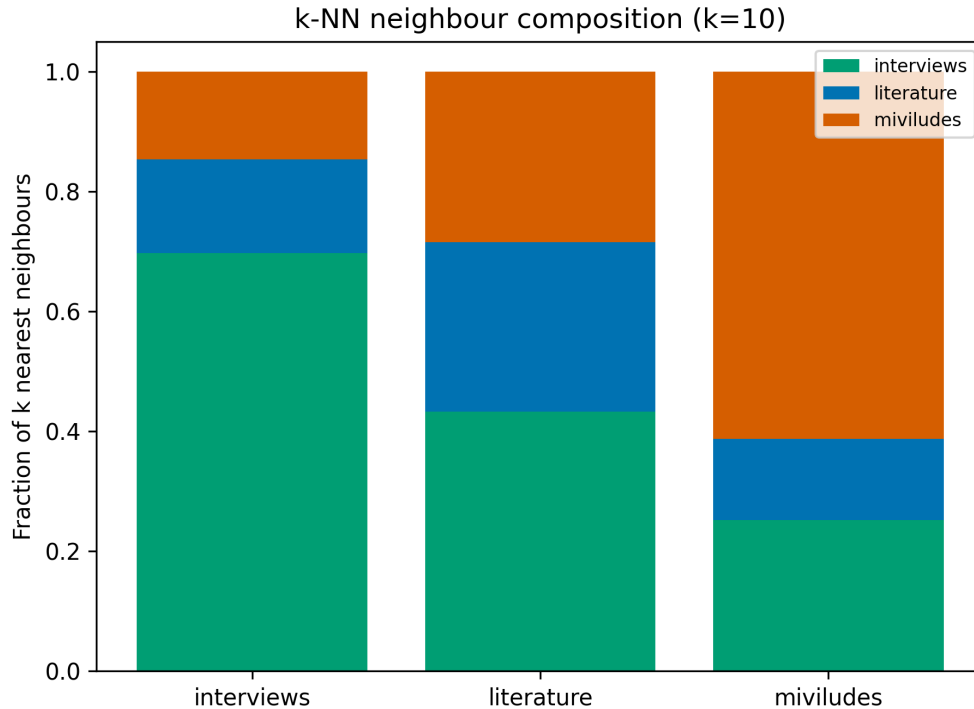


Figure 2: k-NN neighbour composition at $k = 10$, equal- n sampling, v2.

6.4 Criterion neighbours: what sits near MIVILUDES’s 17 official criteria?

Module: `analyze_criterion_neighbours`

Same three-representation design as v1 (French-original, projected English translation, raw English cosine – never merged).

Language sensitivity result. Comparing which corpus is nearest to each of the 17 criteria, French-original versus projected-English-translation (both in the same 389-D shared space): **3 of 17** criteria change which corpus is nearest (v1: **0 of 17**). Comparing against the simpler raw-cosine diagnostic: **4 of 17** differ (v1: **1 of 17**). In plain terms, the choice of language representation matters *more* for v2’s smaller corpus than it did for v1’s – consistent with a smaller candidate pool being more sensitive to exactly which few points happen to be nearest.

6.5 Interview initial exemplars: what do people say first?

Module: `analyze_initial_exemplars`

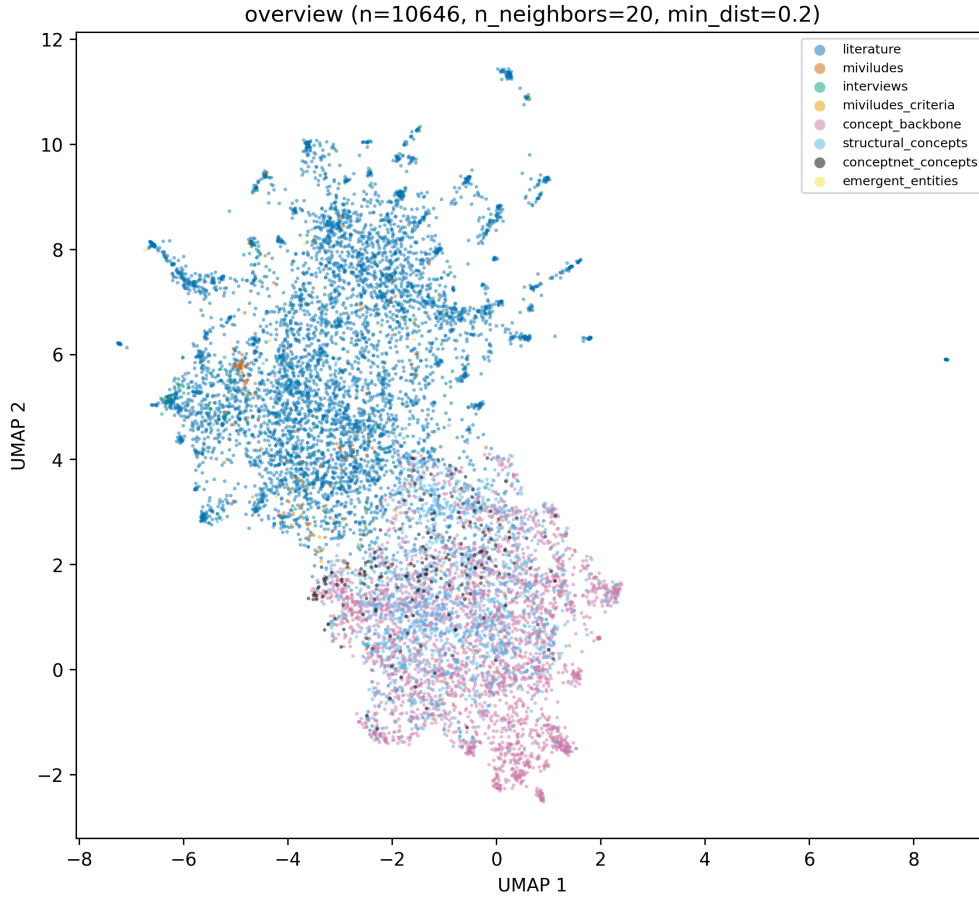


Figure 3: Whole-space UMAP overview, coloured by source dataset. At 10,646 points this is a much less dense picture than v1's 44,520-point equivalent, but is still an exploratory reading aid, not evidence in itself.

The ported prototype layer (Stage 6), unchanged in content from v1 – same 25 resolved / 1 unavailable interviews, same manually-verified quotes – now measured against the v2 space instead of v1's.

Exemplar type	v1 count	v2 count (same quotes)
unclear (incl. 1 unavailable)	12	12
classic_nrm_or_religious_group	4	5
named_group_or_movement	3	3
metaphorical_or_other	2	2
mixed	2	2
digital_or_technology	1	1
organisation_or_institution	1	1

Table 5: Distribution of the 26 interviews' opening responses by descriptive type – unchanged (the underlying quotes were ported, not re-coded); v1's own count for classic_nrm_or_religious_group was listed as 4 with one further "unclear" case since reclassified. Counts shown here are this run's own `exemplar_summary.csv`.

Scope of this evidence. Unchanged from v1: this is exploratory, descriptive evidence from a small

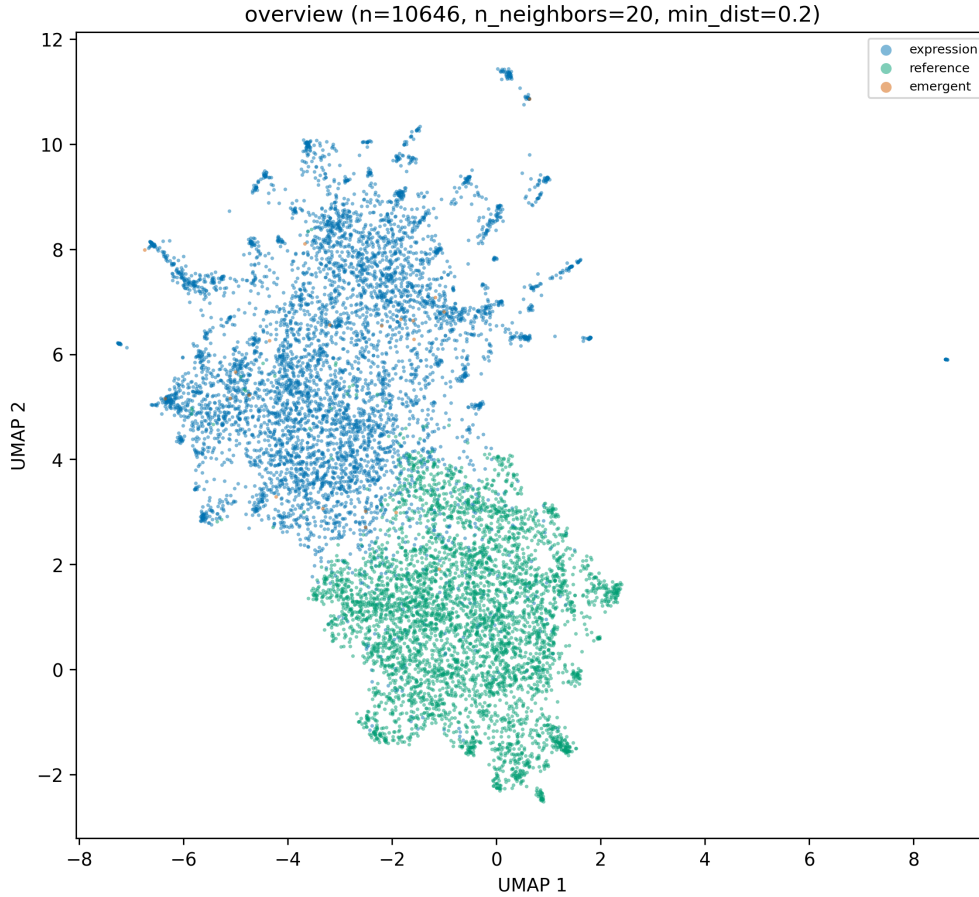


Figure 4: The same UMAP layout, coloured by point role instead of source dataset.

convenience sample – not a representative survey, not a basis for statistical inference. 25 of 26 interviews resolved into a usable geometric position; one had no qualifying opening claim anywhere in its transcript (a genuine absence, not a filtering artefact).

6.6 Emergent entities: what named things does the corpus talk about?

Module: `analyze_emergent_entities`

Provenance category	v1 count (of 3,251)	v2 count (of 21)
literature_only	3,111	20
shared_2_corpora	81	0
miviludes_only	48	1
shared_all_3	9	0
interviews_only	2	0

Table 6: Which corpora mention each emergent entity, v1 versus v2.

A genuinely empty result, not a bug. v2's 21 emergent entities are each mentioned in exactly *one* corpus – zero shared between any two, let alone all three. This toolkit's "shared discourse anchor"

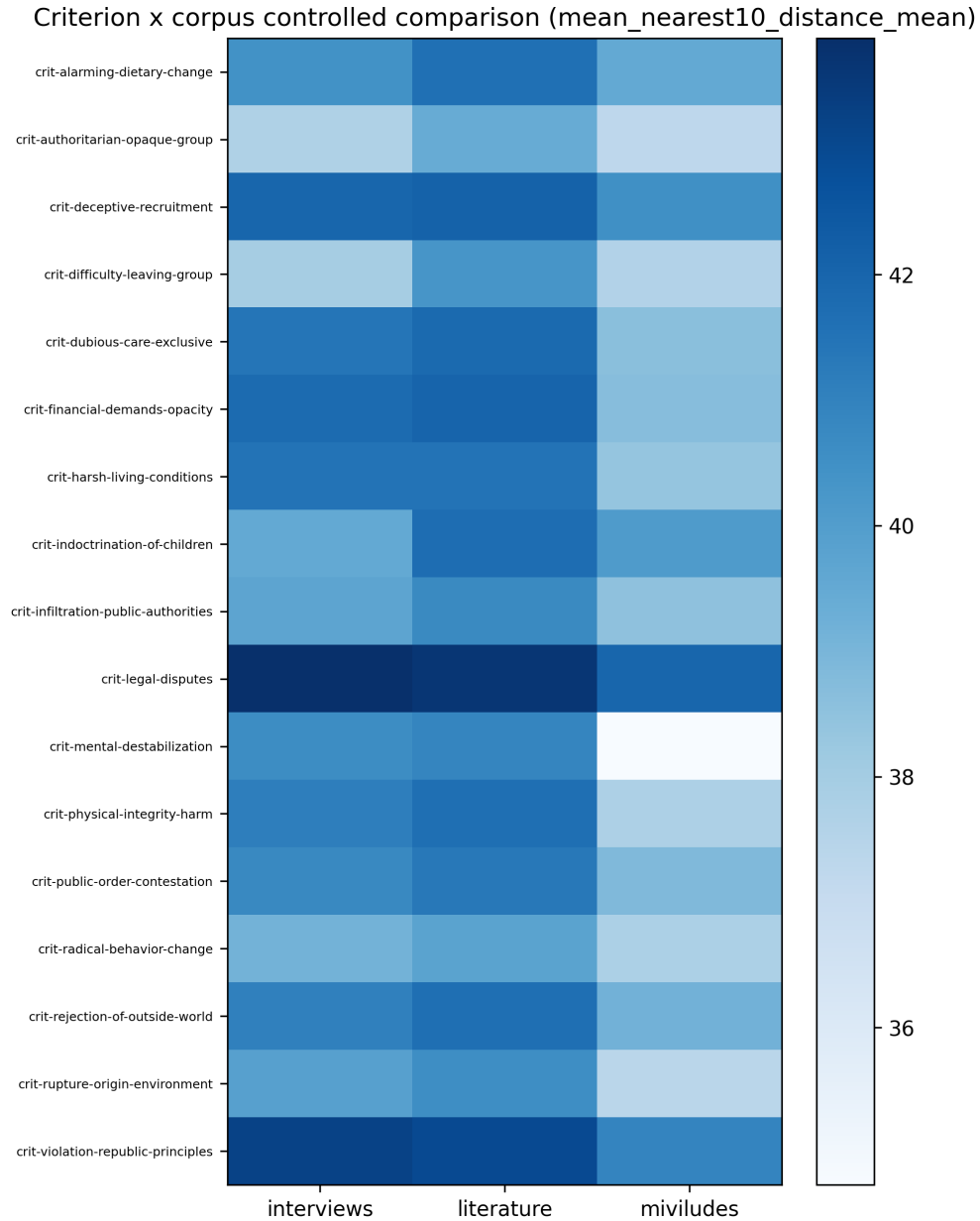


Figure 5: Heatmap of each of the 17 MIVILUDES criteria against the three expression corpora (French-primary representation, v2).

deliverable (retrieving nearest expressions around entities mentioned by all three corpora) is built to fail loudly on an empty result, precisely so an empty result never gets silently treated as "zero anchors found, nothing to see." Direct inspection of every one of the 21 entities' own mention counts confirms this is real: with only 21 entities total (versus v1's 3,251) and interviews contributing just 64 expressions, there simply are not enough independent mentions left for any name to surface in all three sources by chance. The module was adjusted to log this explicitly and skip only that one deliverable, rather than aborting the whole module and losing its other outputs.

As in v1, this module deliberately does **not** classify any entity as "genuinely a cult" versus "mainstream" or apply any other normative label – it reports who talks about what, not who is right.

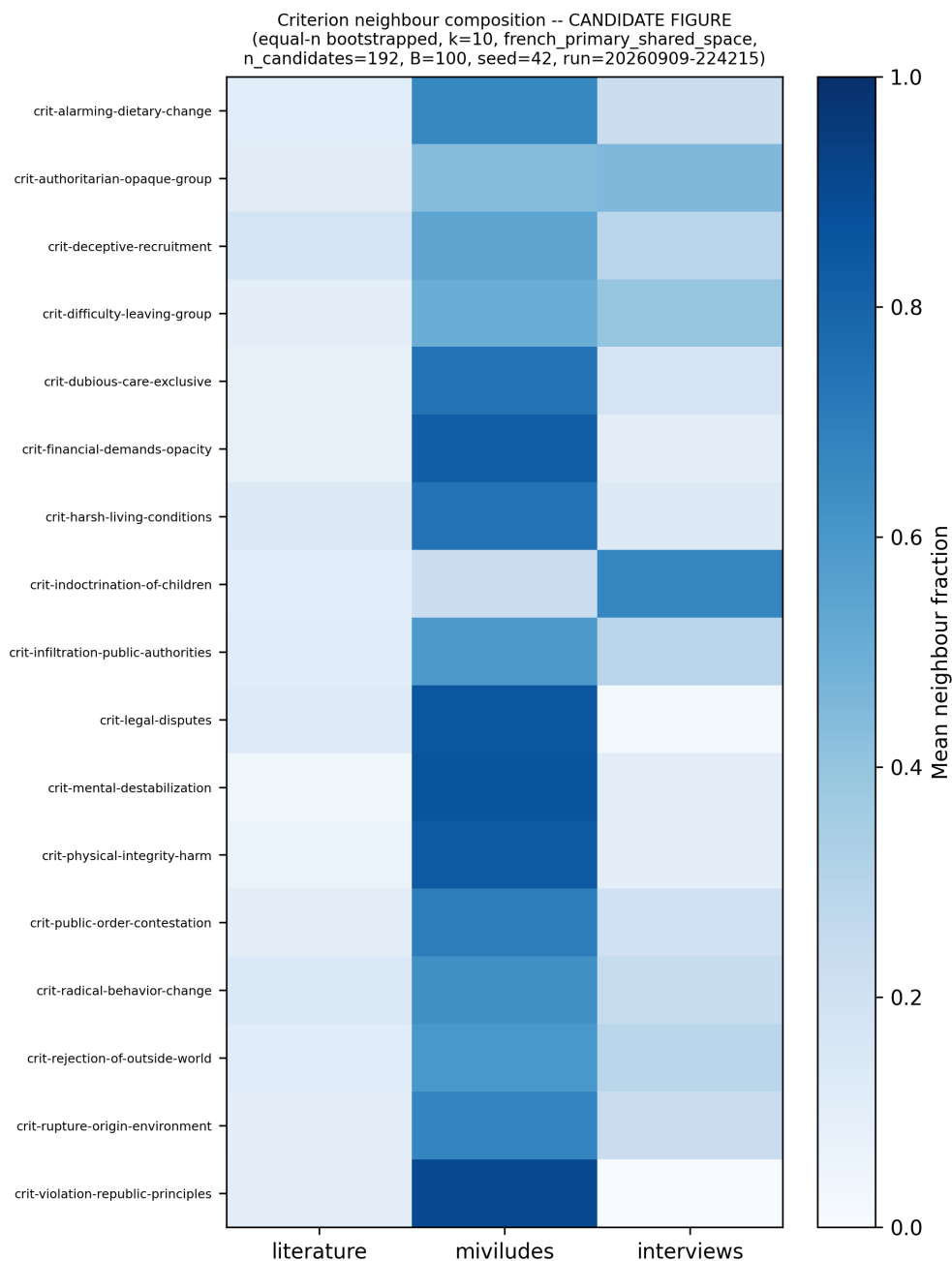


Figure 6: Equal- n , bootstrapped $k=10$ neighbour composition per criterion – a size-controlled companion to the raw heatmap above, new this run.

6.7 Typicality and borderline points (new this run)

Module: `analyze_typicality`

Run for the first time this session (see Section 3.11 for what "typicality" and "borderline" mean here). For each corpus, the 10 points closest to their own corpus centroid ("typical") and the 10 furthest ("atypical") were retrieved; separately, for every pair of corpora (and every pair of epistemic-status groups within a corpus), every point in that pair's pool was ranked by how close its distance to each side's centroid is to being exactly equal ("borderline").

A clear pattern recurs across all three corpora's full top-10 lists (full tables in `typicality_by_corpus.csv`):

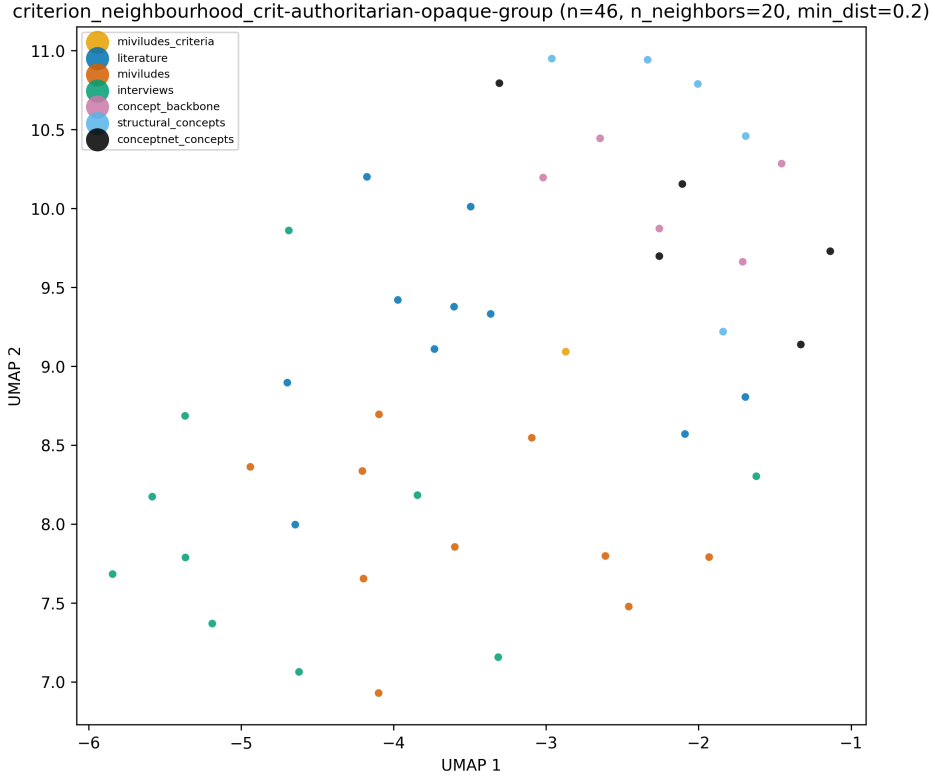


Figure 7: Focused UMAP neighbourhood around the "authoritarian, opaque group" criterion, v2, coloured by source dataset. One of 17 such per-criterion figures generated this run.

Corpus	Most typical (rank 1)	Most atypical (rank 1)
literature	"This is Christianity."	"Jose Silva"
miviludes	"sectarian deviations"	"Scientology"
interviews	"the cult"	"Gourou"

Table 7: The single most typical and most atypical expression per corpus, by distance to that corpus's own centroid.

the **most atypical** items are almost always *specific named things* – a person's name ("Jose Silva," "Asahara"), a specific named group ("Scientology," "Jehovah's Witnesses," "International Church of Good News"), a specific school system ("Steiner-Waldorf schools") – while the **most typical** items are short, generic statements using the corpus's own core vocabulary directly ("sectarian deviations," "the cult," "This is Christianity.," "of a sectarian drift"). (What this might suggest is picked up in Section 8.)

The cross-corpus borderline check found, for example, that the single most-equidistant point between literature and interviews is the phrase "He uses the word cult to 'designate those movements that appear to re[...]'" (distance to both centroids: 30.88, to four decimal places identical) – while between MIVILUDES and interviews, the most equidistant point is a French interview question about "des sectes actuelles" (current-day sects). Full ranked tables (11,826 rows for the three cross-corpus pairs combined) are in `borderline_cross_epistemology.csv`; a matching `borderline_cross_status_within_corpus.csv` covers asserted/speculative/contested pairs within each corpus. No figure accompanies this module this run – these tables read most naturally as tables, not pictures.

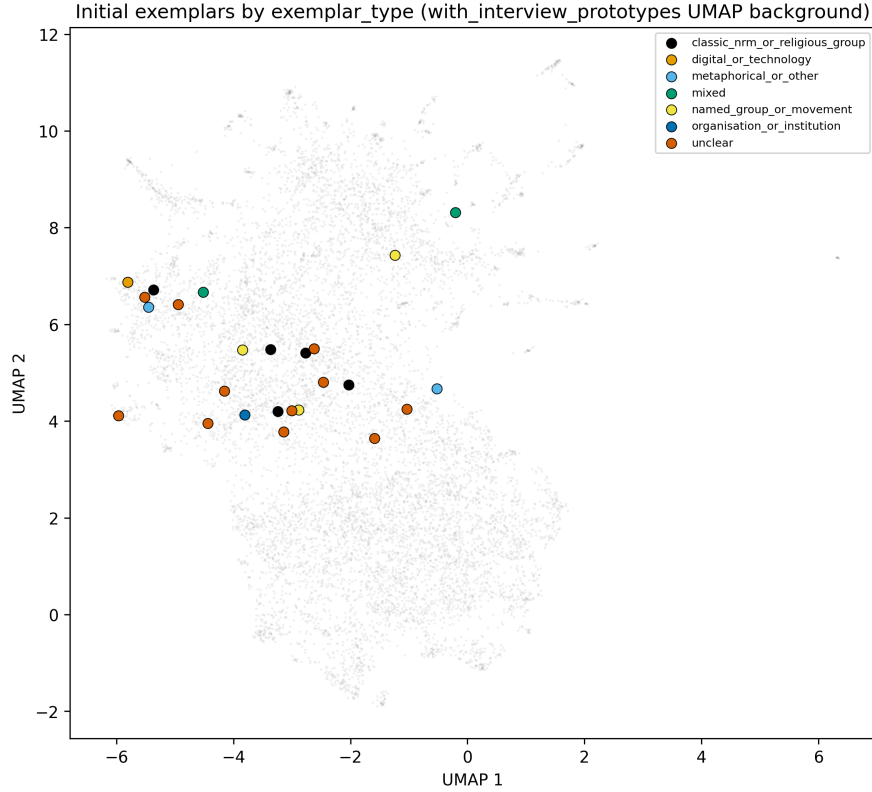


Figure 8: Interview opening responses grouped by exemplar type, v2.

6.8 Focused projections: readable, curated pictures

Module: `generate_focused_projections`

Same design as v1: small, curated point-sets around a specific question (one MIVILUDES criterion’s neighbourhood, the interview prototypes against everything else, the emergent-entity region, ...), each projected both via a fresh UMAP fit and the same fixed PCA basis used everywhere else. 23 such focused populations were generated this run (v1: 22), from 46 points (a single criterion’s neighbourhood) up to 5,095 (the largest reference-vocabulary comparison). Every rendered figure notes `n_points_fit` versus `n_points_overlay` separately; see `generate_focused_projections/config.json` for the full list.

7 Honesty notes – what these numbers do not show

Collected in one place, since they matter for how any of this gets written up. Several are unchanged from the v1 document; new ones are marked.

- **Corpus imbalance persists.** Literature is 97.09% of all v2 expression points (v1: 97.44%) – essentially unchanged, and still requires the equal-*n*/equal-size corrections used throughout.
- **(New) The equal-*n* draw shrank from 204 to 64 points.** The correction logic is unchanged and still necessary, but every bootstrapped confidence interval in this document rests on a smaller, noisier sample than the equivalent v1 number.
- **MIVILUDES is still two documents, now with only 108 expressions surviving strict extraction.** Treat MIVILUDES’s results as one influential operational framework’s

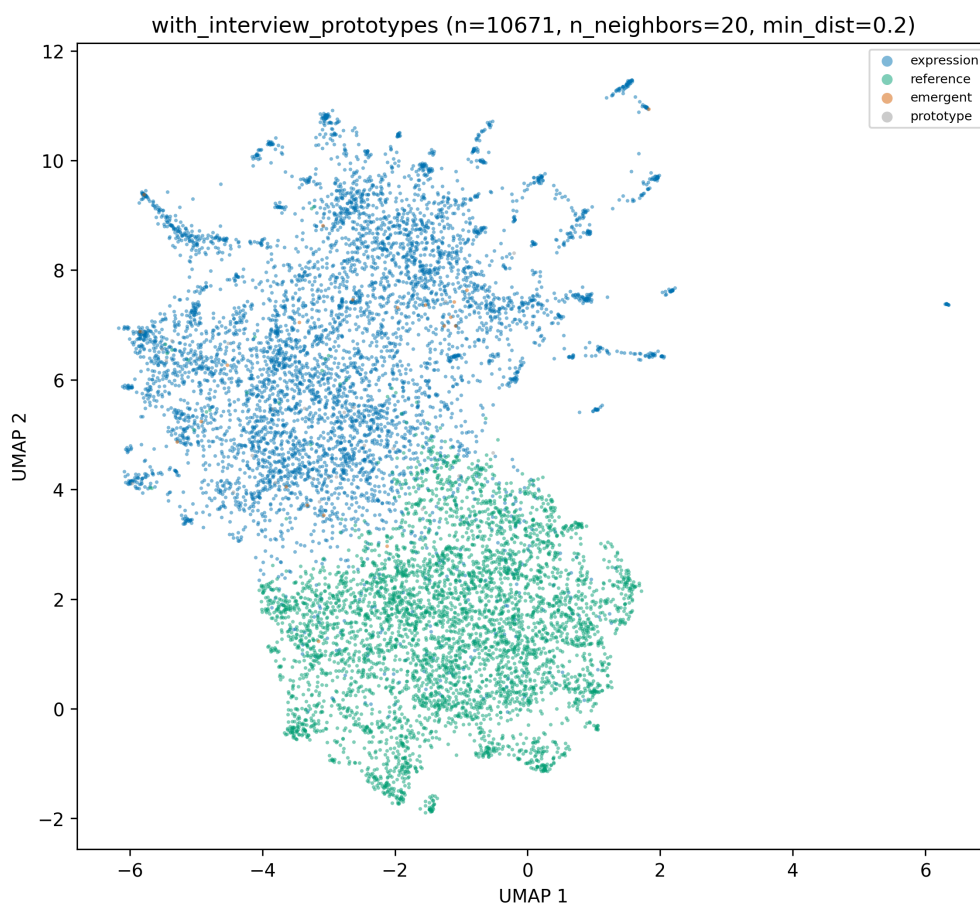


Figure 9: Whole-space v2 UMAP with the 25 resolved interview-prototype points overlaid, coloured by point role.

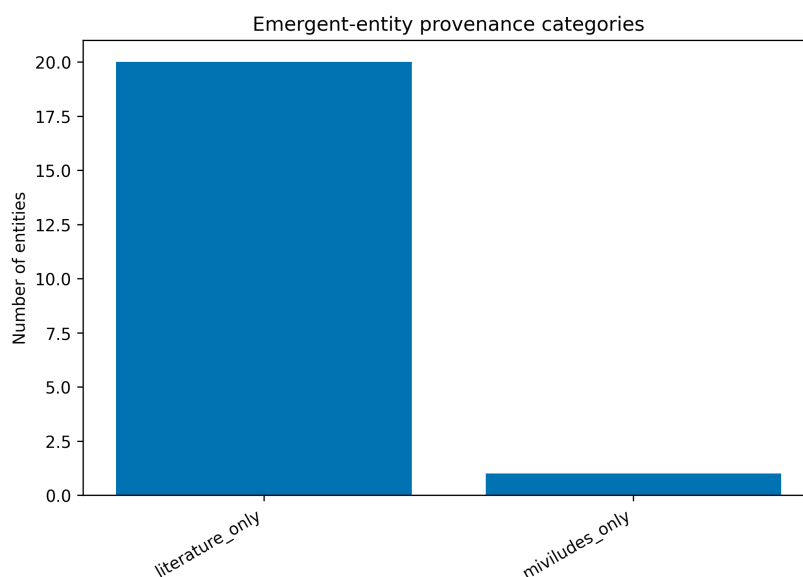


Figure 10: v2 emergent entities grouped by which corpus/corpora mention them.

own text, read very closely – not a representative sample of French state framing in general, and now an even smaller textual base than v1’s already-small one.

- **The interview sample is a convenience sample, now with only 64 surviving expressions.** Exploratory prototype evidence, not a representative survey of lay usage – and now the smallest of the three corpora by a wide margin.
- **response_rank is still not usable** as a "first thing they thought of" signal – only the ported initial-exemplar layer should be used for interview-side "first impression" claims.
- **(New) Zero entities are mentioned by all three corpora at once.** A real structural fact about this run's much smaller corpus (Section 6.6), not a bug and not evidence that the three sources talk about nothing in common – centroid-level and criterion-level comparisons above still find real relationships between the sources even though no single named entity happens to cross all three.
- **(New) One point ("Sept?") carries essentially no propositional content** despite passing every automated quality check (Section 5). Scale is negligible (1/10,646) but the case is real.
- **Equal-size-controlled comparisons control for pool size only.** Still not evidence the three reference vocabularies are otherwise interchangeable.
- **2-D pictures are a reading aid, not evidence.** The real numbers come from the full 389-dimensional space.
- **Three language representations of the MIVILUDES criteria are kept separate on purpose,** and matter *more* for v2 than they did for v1 (Section 6, criterion neighbours) – never averaged together.
- **("New") Typicality/borderline scores are purely geometric,** about this space under this model this run – not a claim about cognitive prototypicality, real-world category membership, or genuine conceptual ambiguity (Section 3.11).

8 Possible interpretations

Read this box before the rest of this section. Everything above this point is descriptive: what was measured, and what came out. Everything in this section is a *reading* of that description – provisional, argued for rather than proven, and offered because it was explicitly asked for as a first pass, not because the geometric evidence alone settles any of it. Each interpretation below is marked with its own tan box and paired with the specific fact it is reading, plus at least one reason it might be wrong. None of this should be copied into the thesis's Results chapter as a conclusion without the researcher independently deciding it holds up.

On the corpora "pointing the same direction" in v2. Section 6's global-structure numbers show cosine similarities between the three corpora's centroids jumping from near-zero/negative in v1 to solidly positive (0.26–0.64) in v2. One reading: v1's over-inclusive extraction pulled in enough noise (paraphrase drift, headings, off-topic bibliography text) that each corpus's average position was itself noisy and only weakly related to what that corpus was actually saying about cults; v2's stricter, verified extraction may be giving a cleaner signal of real overlap in how these three very different sources actually talk about the topic. **Why this might be wrong:** cosine similarity between two centroids is also mechanically sensitive to how many points went into each average and how much they vary – a smaller, more homogeneous sample (v2's) can produce a tidier-looking centroid almost by

construction, regardless of whether the underlying overlap in meaning actually grew. Distinguishing "cleaner signal" from "smaller-sample artefact" would need something this run does not provide, e.g. resampling v1's own corpus down to v2's size and checking whether its cosine similarities move the same way.

On self-clustering dropping for every source in v2. Section 6.3's k-NN table shows every source less self-clustering in v2 than in v1, most sharply for interviews (85.5% → 69.7%). One reading: with a much smaller equal-*n* draw (64 vs. 204 points), each source's own "cloud" is thinner and more easily pierced by a handful of near neighbours from elsewhere, so *some* of this drop is close to a guaranteed side-effect of Stage 7's shrinking sample size, not a genuine change in overlap. A second, not-mutually-exclusive reading: if v1's extraction retained near-duplicate paraphrases and boilerplate within each source (exactly what the fidelity review that motivated v2 found), those duplicates would have artificially inflated *within-source* self-similarity in v1 – so some of the drop could be v1 overstating self-clustering, not v2 understating it. **Why either might be wrong:** both silhouette scores stay near zero regardless, so under either reading the headline conclusion – the three sources substantially overlap rather than forming clean clusters – does not change; what is genuinely uncertain is only the size of the *difference* between v1 and v2 on this specific, sample-size-sensitive statistic.

On named entities sitting at the "atypical" edge. Section 6.7 found, in all three corpora, that the most atypical items (furthest from their own corpus centroid) are specific named people or groups, while the most typical items are generic cult/sect vocabulary. One reading: a corpus centroid is, in effect, an average over a lot of generic descriptive and evaluative language, so it should be geometrically closest to more of that same generic language and furthest from any one idiosyncratic proper name, which by definition carries lexical content specific to itself rather than shared with most other expressions in the corpus. On this reading the pattern says more about how averaging text embeddings behaves than about which named things are "more" or "less" central to cult discourse. **Why this might be wrong, or at least incomplete:** if it were purely a name-versus-generic-language artefact, it should apply equally to *every* named entity regardless of how central or fringe that entity is to the corpus's own concerns – checking whether frequently-discussed named groups (e.g. Scientology, mentioned repeatedly across the literature) are systematically less atypical than one-off names would help separate a pure lexical-averaging effect from one that also tracks how much attention a given named thing actually receives; this run's typicality module was not set up to answer that finer question.

On language-representation sensitivity getting worse in v2. Section 6's criterion-neighbours module found more criteria change their "nearest corpus" answer under a different language representation in v2 (3–4 of 17) than in v1 (0–1 of 17). One reading: with literature down to 5,741 points and MIVILUDES down to 108, the candidate pool feeding each "nearest corpus" decision is much smaller, so a handful of individual points shifting position under a different embedding of the same French text is more likely to actually flip which corpus comes out nearest – a direct, expected consequence of a much smaller corpus, not evidence that v2's translations are somehow less reliable than v1's. **Why this might be wrong:** it is also possible the specific criteria whose ranking flips are ones sitting in a genuinely contested region of the space regardless of corpus size, in which case the increase in flips reflects something substantive about those particular criteria rather than sample size alone – Section 6's

table of which criteria flipped would be the place to check this by re-reading which specific criteria recur across both v1 and v2's flip lists, which this document has not done.

9 Using this document to write the Results chapter

Sections 2–7 are deliberately descriptive, exactly as the v1 document was. Section 8 is new: a first, explicitly provisional pass at interpretation, requested for this iteration specifically, and kept visually and structurally separate from the descriptive material throughout so a reader (or the researcher, coming back to this later) can tell which sentences are geometric fact and which are one possible reading of that fact. It does not itself answer the thesis's actual question – whether the three sources converge on recurring criteria, whether "cult" fragments into genuinely different things across academic, institutional, and everyday discourse, or what kind of category each source implicitly treats it as – and is not a substitute for the researcher independently deciding whether each interpretive box holds up before any of it becomes actual thesis prose.

Concretely, as raw material for that chapter, this document supplies: the same kind of plain-language glossary as the v1 document (now including typicality/borderline); the current v2 numbers for global structure, cluster structure, criterion neighbours, interview exemplars, emergent entities, and typicality, each set directly alongside its v1 counterpart where that comparison is meaningful; a curated set of figures (embedded here, plus the full `generate_figures/` output in `processed/analysis_v2/20260909-224215/` for anything not shown above); an explicit list of what these numbers do not license concluding; and, new this time, a first attempt at interpretation with its own stated reasons it might be wrong – offered as a starting point for argument, not a finished answer.